

Ultrasonic Radar Model Using Microcontroller ATmega128

ASHUTOSH M. BHATT

Radar systems have a number of defence as well as civil applications. Air traffic control uses radars to track aircrafts on the ground and in the air, and to guide planes for smooth landings. Police use radars to detect the speed of passing vehicles. Geologists use radars to map the Earth and other planets. Military uses these for surveillance. Meteorologists use radars to track storms, hurricanes and tornadoes. The list is endless.

A radar system consists of a transmitter that transmits a beam towards the target, which is then reflected by the target as an echo signal. The reflected signal is received by a receiver. This receiver processes

the received signal and provides such information as the presence of a target, distance, position (moving or stationary) or speed, which is displayed on a display unit.

Circuit and its working

Actual radar systems are built with high-power transmitters and receivers, huge antennae, complex processing systems using digital signal processors and large displays.



The circuit described here demonstrates the working of a radar system. It uses ultrasonic waves to detect an object and measure its distance and angular position, and displays the same on a 20x4 LCD screen. It can detect multiple objects at different angles and distances as new objects are detected. This means that the distance and angle of all objects are displayed one by one on the same LCD screen.

The block diagram of an ultrasonic radar system is shown Fig. 1. The system includes an ultrasonic distance measurement (UDM) sensor, LCD panel, opto interrupt sensor, a motor driver, DC motor, buzzer and LEDs. The schematic of the system is shown in Fig. 2.

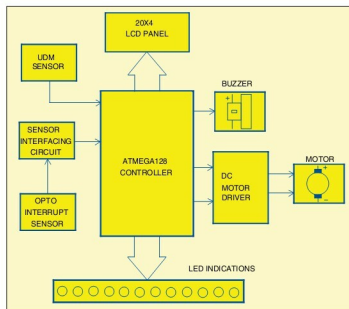


Fig. 1: Block diagram of an ultrasonic radar system

PARTS LIST

Semiconductors:

IC1	- MOC7811 opto isolator
IC2	- L293D DC motor driver
IC3	- ATmega128 microcontroller
LCD1	- 20x4 LCD
T1, T2	- BC547 npn transistor
LED1-LED12	- 5mm LED
LED13	- 5mm

Resistors (all 1/4-watt, 45% carbon):

R1-R12, R19	- 470-ohm
R13, R18	- 330-ohm
R14	- 1k-ohm
R15	- 220-ohm
R16	- 4.7-kilo-ohm
R17	- 10-kilo-ohm
R20	- 100-ohm
VR1	- 10-kilo-ohm-preset

Capacitors:

C1, C2	- 22pF ceramic disk
--------	---------------------

Miscellaneous:

S1	- On/off switch
S2	- Tactile switch
PZ1	- Piezo buzzer
X _{cr} 1	- 12MHz crystal oscillator
CON1, CON4	- 2-pin terminal connector
CON2	- 4-pin connector
CON3	- 3-pin connector
CON5	- 2-pin connector
	- 12V-5RPM DC motor
	- 3-pin ultrasonic distance sensor—serial out

TABLE I
UDM Data Frame Format

byte	Position	Hex value	Description
1	Xxx.xcmm<nl>	0x30 – 0x39	Hundred character
2	xXx.xcmm<nl>	0x30 – 0x39	Tens character
3	xxX.xcmm<nl>	0x30 – 0x39	Ones character
4	.	0x2E fixed	Dot character
5	xxxX.xcmm<nl>	0x30 – 0x39	1/10 character
6	xxxX.xcmm<nl>	0x30 – 0x39	1/100 character
7	c	0x63 fixed	Character c
8	m	0x6D fixed	Character m
9	<new line>	0x0D fixed	New line character

Description of components

Microcontroller (ATmega128). AVR microcontroller ATmega128 (IC3) is a high-performance, low-power Atmel 8-bit AVR RISC based microcontroller that combines 128kB of